

Lab 7

(Constructor + Class)

Q1: Problem Statement: Create a class named Student with instance variables name and marks. Use a constructor to initialize these values. Include a method calculateGrade() that returns "A" if marks ≥ 90 , "B" if ≥ 75 , and "C" otherwise.

Q2: Create a class Employee with id, salary, and performanceScore. Use a constructor to initialize them. Add a method calculateBonus() that adds 10% to the salary if the performanceScore is greater than 8.0, otherwise adds 5%

Q3: Create a class Student with attributes name and rollNumber. Use a constructor to initialize these values when an object is created. Add a method displayInfo() to print the details of the student.

Q4: Create a Student class. Use Scanner in the main method to take the student's name and marks, then use the constructor to initialize the object and display the grade.

Q5: Create a class Calculator. Use a constructor to take two numbers from the user and a method add() to show the sum.

Q6: Create a class Circle. Take the radius as input from the user, initialize it via constructor, and calculate the area ($\text{Area} = 3.14 * r * r$).